



AWS Glue and Google Cloud Equivalents

The 2026 data integration landscape, side by side

TechCatalyst Data Engineering 2026 | Week 3 Day 5 | AWS Glue ETL

CONTEXT

Why this comparison matters

- You just built two Glue Visual ETL jobs. Now we zoom out to the whole platform.
- AWS packages data integration as one product. Google Cloud spreads it across a portfolio.
- Same capabilities, different packaging: the architecture skill is knowing what maps to what.
- Picking the right abstraction level, from no-code to full code, is a design decision, not a default.

1

integrated AWS service

vs 7+

purpose-built GCP services

LEARNING OBJECTIVES

What you will be able to do

- 1 Describe the major AWS Glue components and the 2026 runtime (Glue 5.1, Spark 3.5.6).
- 2 Compare the five Glue authoring modes and choose the right one for a given persona.
- 3 Map each Glue capability to its Google Cloud equivalent service.
- 4 Decide between Cloud Dataflow and Managed Service for Apache Spark for a workload.
- 5 Explain why Apache Iceberg and generative AI are reshaping both platforms.

PART I: AWS GLUE

Glue is a platform, not a single tool



Serverless: no clusters to manage 1 DPU = 4 vCPU + 16 GB **\$0.44 per DPU-hour (FLEX: \$0.29)**

PART I: AWS GLUE

Five terms you need in your vocabulary



Data Catalog

One per region. Databases, tables, partitions, connections, statistics. Athena, Redshift, and EMR all read it.



DynamicFrame

Glue's native data structure. Each record carries its own schema, so schema drift and messy nested data do not break the job.



Crawler

Connects to S3, JDBC, or DynamoDB, infers schemas with classifiers, and auto-populates catalog tables.



Job + Trigger

A job is the execution unit (script, sources, targets). Triggers start it: cron, on-demand, events, or job chaining.

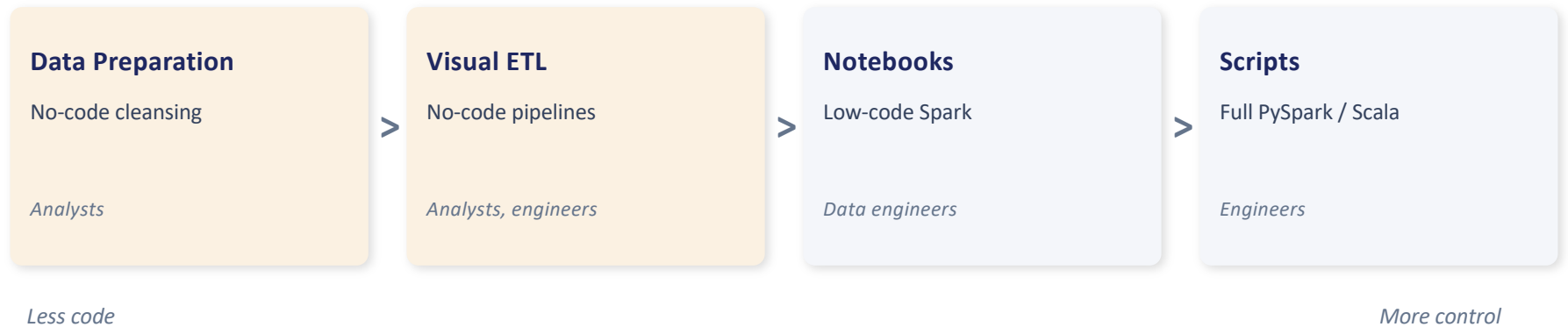


Workflow

A DAG of crawlers, jobs, and triggers. Lightweight orchestration without Airflow.

PART I: AWS GLUE

Five authoring modes, one platform



Amazon Q Data Integration spans all four

Describe a pipeline in plain English and Q generates the visual DAG or the PySpark code. A fifth modality layered on top of the others.

No-code: Glue Studio Visual ETL

- Drag-and-drop canvas: sources, transforms, and targets as connected nodes.
- Generates real PySpark underneath. View it, copy it, keep learning from it.
- Data preview shows live sample rows at every node while you build.
- Transform palette: filter, join, aggregate, union, SQL node, custom Python node, PII detection.
- 2025 to 2026: Excel sources, single-file output, more compression codecs, and Amazon Q chat that builds the DAG for you.



You used this today

Your bronze ingest and silver transform jobs are Visual ETL jobs. Everything on this slide should feel familiar.

No-code: Data Preparation (formerly DataBrew)

- Spreadsheet-style grid: click a column, pick a transformation, see the result.
- 250+ ready-made transforms: fix nulls, standardize dates, validate emails, mask PII.
- Steps are saved as recipes: versioned, reusable sequences you can apply to any dataset.
- Built-in profiling: column stats and distributions before you clean.
- Absorbed into Glue Studio in July 2024; existing DataBrew recipes import directly.

Which no-code tool?

Data Preparation

Cleansing one dataset, quality fixes, exploration.

Visual ETL

Multi-source pipelines, joins, loading lakes and warehouses.

Low-code: Studio Notebooks on Interactive Sessions

- Jupyter notebooks backed by serverless Spark that starts in seconds.
- PySpark only in the console; full DataFrame and Dataset APIs.
- Same engine behind the data preview pane you used in the lab.
- Open kernel works in VS Code and PyCharm; May 2026 adds native Spark Connect for any remote client.
- SageMaker Unified Studio can now use Glue 5.1 as its notebook Spark runtime.

Session magics

```
%glue_version 5.1
%worker_type G.1X
%number_of_workers 2
%additional_python_modules
%idle_timeout 30
%stop_session
```

Code-first: PySpark and Scala scripts

```
from awsglue.context import GlueContext
glueContext = GlueContext(sc)

dyf = glueContext.create_dynamic_frame \
    .from_catalog(database="my_db",
                  table_name="my_table")

clean = DropNullFields.apply(dyf)

glueContext.write_dynamic_frame \
    .from_options(frame=clean,
                  connection_type="s3",
                  connection_options={"path": "s3://out/"},
                  format="parquet")
```

Worker types (pick your muscle)

Type	Specs	Use
G.025X	2 vCPU, 4 GB	24/7 streaming
G.1X	4 vCPU, 16 GB	Default batch
G.2X to G.16X	up to 256 GB	Wide joins, ML
R series	2x memory	Memory heavy

Auto Scaling adjusts worker count mid-run. Your lab jobs pinned 2 x G.1X on purpose.

AI-assisted: Amazon Q Data Integration



Generate

Describe the pipeline in English, receive a runnable Glue job with config.



Refine visually

Chat adds, edits, or removes nodes on the Visual ETL canvas, multi-turn.



Troubleshoot

Reads run logs and metadata, returns root cause and a fix, minutes not days.



Upgrade Spark

Finds API and behavior changes when moving jobs from Spark 2.4 to 3.5.

Code generation is PySpark only (no Scala). Available in the Glue console and SageMaker Unified Studio.

PART I: AWS GLUE

The 2026 runtime: Glue 5.1

Version	Spark	Python	Table formats
Glue 4.0	3.3.0	3.10	Iceberg 1.0, Delta 2.2, Hudi 0.12
Glue 5.0	3.5.2	3.11	Iceberg 1.6, Delta 3.1, Hudi 0.15
Glue 5.1 (current)	3.5.6	3.11	Iceberg 1.10, Delta 3.3.2, Hudi 1.0.2

Why Glue 5.1 matters

- Apache Iceberg format v3: deletion vectors, default column values, row lineage.
- Lake Formation fine-grained access control now applies to writes, not just reads.
- Faster S3 access (S3A default connector) and a built-in Spark troubleshooting agent.

Beyond batch: the rest of the platform



Streaming ETL

Kinesis, MSK, and Kafka sources with Spark Structured Streaming, checkpointing to S3, and the tiny G.025X worker for always-on jobs.



Data Quality

DQDL rules on the Deequ engine, 25+ rule types, inline pipeline gates, and ML anomaly detection that writes rules for you.



Zero-ETL

Managed replication: DynamoDB, Salesforce, SAP, plus self-managed Oracle, SQL Server, MySQL, and PostgreSQL into Redshift or S3 Tables.



Schema Registry

Free, serverless registry for Avro, JSON, and Protobuf. Blocks breaking schema changes before they reach consumers.

Governance: the catalog plus Lake Formation

- Lake Formation builds its permission layer directly on the Glue Data Catalog.
- Fine-grained control: row-level, column-level, and cell-level security on tables.
- Glue 5.1 extends fine-grained control to write operations in Spark.
- Tag-based access control: policy by metadata tag instead of per-table grants.
- Cross-account sharing for data mesh patterns.



New frontier: catalog federation

Since Nov 2025, Glue federates to any Iceberg REST catalog, including Databricks Unity Catalog and Snowflake Horizon. Query remote tables without moving data, Lake Formation permissions intact.

PART II: GOOGLE CLOUD

GCP has no Glue

The same capabilities exist, distributed across a family of purpose-built services. You are the integrator.



AWS Glue

One service umbrella: catalog, ETL, streaming, quality, replication, AI assist

Cloud Data Fusion

Dataflow

Managed Spark

Dataform

Datastream

Knowledge Catalog

+ BigQuery Pipelines, Pub/Sub Schema Registry, Cloud Composer for orchestration

PART II: GOOGLE CLOUD

The capability map

AWS Glue capability	GCP equivalent
Visual no-code ETL	Cloud Data Fusion (Pipeline Studio + Wrangler)
PySpark / Scala batch ETL	Managed Service for Apache Spark (formerly Dataproc)
Unified batch + streaming code ETL	Cloud Dataflow (Apache Beam)
Interactive Spark notebooks	Managed Spark interactive sessions
SQL transformation pipelines	Dataform (inside BigQuery Studio)
Zero-ETL / CDC replication	Datastream
Data catalog and governance	Knowledge Catalog
Data quality	Knowledge Catalog data quality checks
Workflows / orchestration	Cloud Composer (Airflow) or BigQuery Pipelines
Schema registry	Pub/Sub Schema Registry
GenAI assistance (Amazon Q)	Gemini in BigQuery and Data Fusion

Cloud Data Fusion: the visual ETL hub

- Fully managed visual integration built on open-source CDAP: pipelines are portable.
- Pipeline Studio: node-based canvas with 100+ source, transform, and sink plugins.
- Wrangler: spreadsheet-style cleansing, the Data Preparation counterpart.
- Runs each pipeline on ephemeral Spark clusters it provisions for you.
- Built-in replication template gives no-code CDC from Oracle into BigQuery.

Watch the meter

Developer: ~\$1.85/hr

Basic: ~\$9.60/hr + compute

Enterprise: HA + security

Instances bill while running, unlike Glue jobs that bill per run. The biggest cost-model difference between the two.

Two engines for code: Managed Spark vs Dataflow

April 2026: Dataproc and its serverless mode were unified as Managed Service for Apache Spark. Dataflow remains the Apache Beam service with one model for batch and streaming.

Factor	Choose Dataflow	Choose Managed Spark
Programming model	Greenfield, Apache Beam	Existing Spark or Hadoop code
Streaming	First-class, unified with batch	Possible, but secondary
Infrastructure control	Minimal, fully serverless	High: clusters, images, SSH
Ecosystem	Beam IO connectors	Any Spark library, Hive, Pig
Cost model	Pay only while processing	Cluster-based or per-job serverless
2026 edge	Liquid sharding, TPU workers	Spark 3.5 runtime, Glue-like sessions

The rest of the family



Datastream

Serverless log-based CDC from MySQL, PostgreSQL, Oracle, and SQL Server into BigQuery and Iceberg. Handles schema drift with a versioned registry. 500B+ events per month.



Dataform

SQL-first pipelines with SQLX: ref() builds the DAG, assertions test quality, Git is native. Free; you pay only BigQuery compute. The dbt-style layer.



Knowledge Catalog

Universal catalog and governance: auto-discovery, business glossary, column-level lineage, and Gemini-generated descriptions across the estate.



BigQuery Pipelines

Sequence SQL, notebooks, and AI-assisted data preparation inside BigQuery Studio. Gemini turns plain-language requests into transformation steps.

HEAD TO HEAD

Authoring modes, both clouds

Mode	AWS	Google Cloud
No-code visual canvas	Glue Studio Visual ETL	Data Fusion Pipeline Studio
No-code cleansing	Glue Data Preparation	Wrangler / BigQuery data preparation
Interactive notebooks	Glue Interactive Sessions	Managed Spark sessions
Code-first Spark	Glue ETL scripts	Managed Spark (serverless or cluster)
Code-first Beam	Not offered	Cloud Dataflow
SQL pipelines	Spark SQL in Glue jobs	Dataform in BigQuery Studio
CDC / replication	Glue Zero-ETL	Datastream
GenAI authoring	Amazon Q Data Integration	Gemini in BigQuery / Data Fusion

Key takeaways

1

Platform vs portfolio

Glue is one umbrella; GCP makes you the architect who assembles the pieces.

2

Pick abstraction deliberately

No-code to full code is a continuum. Choose by persona and use case, not prestige.

3

Iceberg is the currency

Both clouds standardized on Apache Iceberg. Table formats now outrank engines.

4

GenAI is production-grade

Amazon Q and Gemini author real pipelines. Prompting joins the curriculum.

5

Catalogs are federating

Iceberg REST lets Glue query Unity and Horizon catalogs in place. Data mesh gets real.

6

Judgment beats tooling

You debugged raw parquet this week. No canvas replaces knowing what is underneath.

WRAP UP

Where this goes next

- Finish the lab deliverables: bronze and silver evidence plus your medallion notes.
- Week 4: Snowflake and advanced SQL. Watch for the same medallion vocabulary.
- Week 5: you write the PySpark yourself. Today's generated scripts are your preview.
- Discussion prompt: your team inherits a GCP estate after an acquisition. Which three services replace your Glue pipelines, and what breaks?

Go deeper

[AWS Glue documentation](#)

[Glue 5.1 release notes](#)

[Cloud Data Fusion docs](#)

[Dataflow innovations blog](#)

[Dataform SQLX guide](#)

Full comparison guide in the lab folder